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&
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PRYON

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America's AI Dominance: Efficient, Secure, and Sovereign

Recommendations for the Artificial Intelligence Action Plan

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I. Executive Summary

i. Overview

Artificial intelligence stands as a cornerstone of American leadership in the 21st century, promising to enhance human flourishing, economic competitiveness, and national security as outlined in [Executive Order 14179](#) (January 23, 2025). The Trump Administration’s revocation of burdensome regulations under [EO 14110](#) and its call for an AI Action Plan signal a pivotal shift toward unleashing private sector innovation. This white paper responds to the Office of Science and Technology Policy’s (OSTP) solicitation by addressing critical AI policy topics and proposing actionable policies to solidify U.S. dominance, guided by economic and security imperatives. Current AI challenges, such as resource-intensive models, security vulnerabilities, and workforce inefficiencies, demand solutions that are compute-efficient, secure, and scalable. The recommendations herein reflect a vision for responsible AI that empowers federal agencies and the private sector alike, ensuring America leads without ideological overreach, engineered social agendas, or unnecessary bureaucracy.

ii. Respondents

This white paper is submitted jointly by Pryon Inc. and Defense Industry Advisors, two organizations uniquely equipped to inform the AI Action Plan under Executive Order 14179.

Pryon Inc., a leader in AI-driven knowledge management, delivers a scalable, SOC 2-compliant Retrieval-Augmented Generation (RAG) Suite that empowers federal agencies with efficient, secure, and adaptable solutions. By providing natural language answers with attribution from over one million documents, Pryon reduces knowledge friction, enhances workforce productivity, and leverages existing infrastructure, making it a proven partner for federal innovation, as evidenced by deployments with the Air Force, IRS-CI, and intelligence communities.

Defense Industry Advisors (DIA) is a Certified Veteran-Owned Business with over 75 years of combined experience in government acquisitions, DoD manufacturing, and operational execution. DIA is already working to improve government efficiency by developing AI algorithms that aim to streamline acquisition and diversify the industrial base.

Together, Pryon and Defense Industry Advisors offer a trusted blend of cutting-edge AI technology and deep defense market insight, positioning them as authoritative voices to shape America’s AI leadership strategy.

iii. Policy Recommendations

The following policies align with the president’s ‘National Artificial Intelligence Research and Development Strategic Plan’ as well as the White House’s ‘Blueprint for an AI Bill of Rights’.

- Accelerate workforce efficiency with rapid-deployment AI.
- Mandate compute-efficient AI standards for sustainability.
- Enhance open-source utility with secure hybrid frameworks.
- Strengthen cybersecurity with sovereign AI deployments.
- Safeguard data privacy with pretrained, no-training AI.
- Streamline governance with flexible, transparent standards.

II. Application and Use

Federal agencies and private enterprises increasingly rely on AI to streamline operations and enhance decision-making, yet many solutions fail to integrate seamlessly with existing workflows or adapt to dynamic needs. Knowledge friction, delays in accessing critical information, plagues workforces, particularly as employee turnover rises and documentation sprawls across decentralized systems. Existing AI applications often prioritize novelty over practicality, leaving users with tools that are slow to deploy or misaligned with real-world demands, risking the safe and effective systems principle from the [Blueprint for an AI Bill of Rights](#).

The future lies in AI systems that deliver immediate, attributable answers from trusted sources, reducing onboarding time and augmenting human capabilities. Such tools should span data silos, integrate with platforms like SharePoint or ServiceNow, and support multimodal content e.g., handwritten notes, embedded images, to maximize utility across sectors. These systems must provide Notice and Explanation, clear documentation of their function and impact, ensuring users understand how automation affects them. This approach aligns with the [Select Committee on Artificial Intelligence](#)'s Strategy 2, human-AI collaboration, and Strategy 7, workforce readiness, while meeting the Blueprint's call for unified access to critical resources.

III. Energy Consumption and Efficiency

The AI boom has driven an insatiable demand for compute power, straining energy resources and inflating costs. Traditional large language models (LLMs), with their massive parameter counts, consume excessive energy for training and inference, posing sustainability challenges for data centers and edge deployments alike. This inefficiency hampers scalability and burdens federal budgets, undermining economic competitiveness and risking unsafe outcomes contrary to a safe and effective systems principle.

Lightweight, compute-efficient AI models that achieve high accuracy with minimal resource demands are essential. Solutions that reduce LLM size by orders of magnitude, while maintaining performance and undergoing pre-deployment testing offer a path to sustainable innovation. This approach aligns with the Select Committee's Strategy 1, long-term investment in responsible AI, and EO 14179's barrier-removal ethos, ensuring systems are both effective and environmentally responsible.

IV. Open-Source Development

Open-source AI has fueled rapid advancements, but it often comes with trade-offs: lack of enterprise-grade security, inconsistent performance, and reliance on fragmented toolsets requiring extensive customization. Federal adoption lags as open-source solutions struggle to meet stringent compliance needs e.g., IL5/IL6, leaving agencies dependent on costly bespoke development.

A hybrid approach, combining open-source flexibility with proprietary optimizations, can accelerate deployment while ensuring reliability. Pretrained, no-training-required models that integrate with existing infrastructure and undergo independent evaluation bridge this gap. This aligns with the Select Committee's Strategy 8, public-private partnerships, and the Blueprint's emphasis on proactive risk mitigation, transitioning innovations into practical, safe use.

V. Cybersecurity

AI systems remain vulnerable to adversarial attacks, data breaches, and supply chain risks, particularly when reliant on external APIs or cloud-hosted models. Federal cybersecurity standards (e.g., NIST 800-53) demand robust protections, yet many commercial offerings fall short, exposing sensitive operations to exploitation. The proliferation of generative AI amplifies risks like hallucination and misinformation.

Secure AI must operate within firewalls, eliminate external dependencies, and incorporate role-based access controls (RBAC). Systems designed for on-premises or air-gapped deployment, with pre-vetted connectors and continuous monitoring can meet these needs. This approach aligns with the Select Committee's Strategy 4, safety and security, and EO 14179's national security focus, ensuring AI safeguards civil liberties and critical operations.

VI. Data Privacy and Lifecycle Security

Data privacy concerns loom large as AI systems ingest vast datasets, often training on customer data and risking leakage or reverse-engineering. Lifecycle security, from development to deployment, is inconsistent, with many solutions lacking granular access controls or auditability. This erodes trust and compliance with federal mandates, conflicting with the Blueprint's Data Privacy principle, which demands built-in protections and user agency.

AI platforms that never train on customer data, maintain read-only access to repositories, and log every transaction for governance offer a solution. Such systems, designed with privacy-by-default features and enhanced protections for sensitive domains e.g., health, education, ensure privacy and security throughout their lifecycle. This aligns with the Select Committee's Strategy 3, ethical and legal implications, and EO 14179's emphasis on sovereignty.

VII. Risks, Regulation, and Governance

The National Institute of Standards and Technology's [Artificial Intelligence Risk Management Framework \(AI RMF 1.0\)](#) highlights that AI's rapid evolution outpaces regulatory frameworks, leading to inconsistent standards. Risks like bias, hallucination, and unintended outputs persist, while overregulation, such as the repealed EO 14110, stifles innovation. Governance struggles to balance accountability with agility, particularly in federal contexts requiring transparency and compliance, echoing the Blueprint's call for Algorithmic Discrimination Protections and Human Alternatives, Consideration, and Fallback.

Flexible, risk-mitigating AI frameworks that ground responses in authoritative content, provide dashboards for usage oversight, and ensure human escalation options can address these challenges. Lightweight regulation that encourages innovation while enforcing security and accuracy via proactive disparity testing and plain language reporting aligns with the Select Committee's Strategy 6, standards and benchmarks, and EO 14179's deregulatory stance.

VIII. Policy Recommendations

The following policy actions address the current state of AI and position the U.S. as the global leader, integrating the *Blueprint for an AI Bill of Rights* to ensure responsible innovation:

1. Accelerate Workforce Efficiency with Rapid-Deployment AI

- o *Action:* Establish a federal AI adoption program to deploy lightweight, pretrained AI

tools across 50% of agencies by 2026, prioritizing solutions that integrate with existing systems and deliver attributable answers in under two weeks, with clear Notice and Explanation documentation.

- o *Rationale:* Knowledge friction hampers productivity; transparent, efficient AI augments human work, supporting Strategy 2, EO 14179's competitiveness goal, and the Blueprint's user-awareness principle.
- o *Impact:* Reduces onboarding burdens and enhances decision-making, mirroring federal successes.
- o *Proposed Policies in Action:* Empower employees with an AI chatbot, such as 'The Pryon Chatbot', that is directly connected with relevant content to: answer inquiries on expense policies, auto-develop daily intel briefing memos, or address IT issues and submit helpdesk tickets.

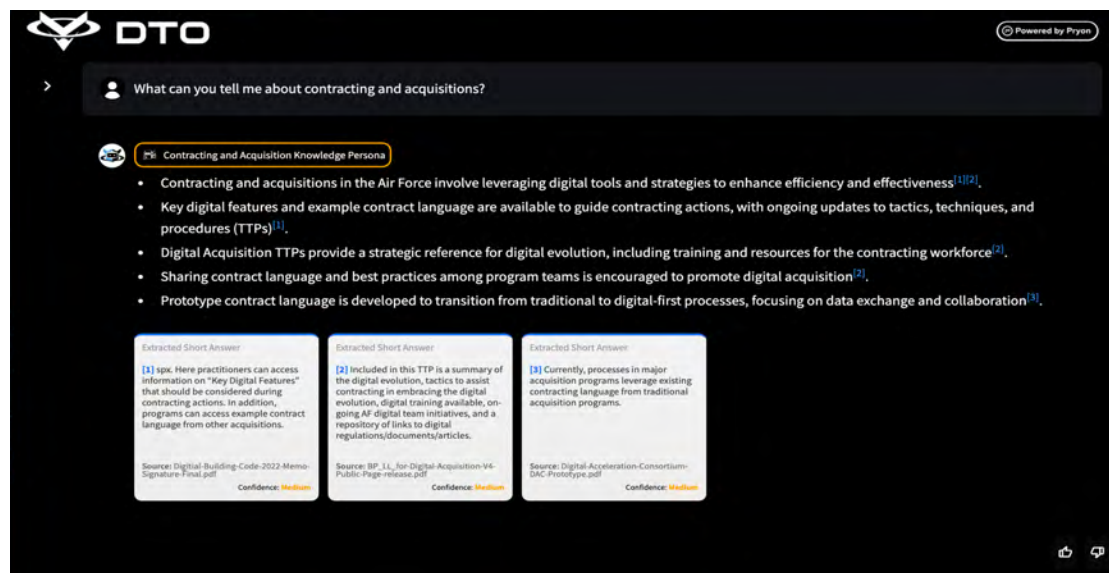


Figure: Pryon AI Chatbot on Air Force Digital Transformation Office site:

<https://dafdto.com/>

2. Mandate Compute-Efficient AI Standards for Sustainability

- o *Action:* Require federal AI procurements to prioritize models with at least 90% reduced compute footprints by 2027, funding R&D to achieve this through public-private collaboration, with pre-deployment testing per Safe and Effective Systems.
- o *Rationale:* Energy-intensive LLMs strain infrastructure; efficient, tested alternatives align with Strategy 1, EO 14179's innovation focus, and the Blueprint's safety mandate.
- o *Impact:* Lowers TCO and scales AI sustainably.
- o *Proposed Policies in Action:* "Eco-scaling" techniques offer various methods to optimize energy consumption during ingestion, inference, and model training processes. For example, when training natural language processing models, Pryon's generative AI platform consumes up to 70% less energy compared to traditional model training approaches. Additionally, end-to-end platform solutions are more efficient and cost effective as compared to integrating multiple, disparate components.

3. Enhance Open-Source Utility with Secure Hybrid Frameworks

- o *Action:* Launch an AI Innovation Hub to blend open-source flexibility with enterprise-grade security features, e.g., no-training models, IL6/IL7 compliance, by 2026, ensuring

independent evaluation per the Blueprint.

- o *Rationale:* Open-source lacks federal readiness; hybrid systems support Strategy 8, EO 14179's private sector empowerment, and Safe and Effective Systems.
- o *Impact:* Speeds innovation with reliability.
- o *Proposed Policies in Action:* Agencies are advised to utilize retrieval-augmented-generative (RAG) AI solutions, like Pryon AI, which leverage closed-source LLMs with source attribution for improved accuracy but also provide the flexibility to connect to open-domain LLMs when stringent regulatory compliance is not required.



Figure: Pryon ingestion, retrieval, and generative engines utilizing local LLM and attributions to content source.

4. Strengthen Cybersecurity with Sovereign AI Deployments

- o *Action:* Mandate that AI systems handling sensitive data operate on-premises or air-gapped, with no external API reliance, achieving IL6 compliance by 2026, and continuous monitoring per the Blueprint.
- o *Rationale:* Vulnerabilities threaten security; self-contained AI aligns with Strategy 4, EO 14179's security priority, and Data Privacy.
- o *Impact:* Protects critical operations.
- o *Proposed Policies in Action:* On-Premise AI deployment, preferably air-gapped, with read-only, connections which respect the inherent user permissions to content, is recommended for IL-6+ compliance.

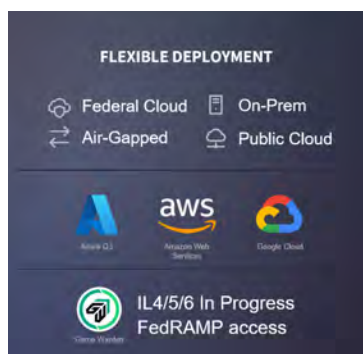


Figure: Pryon secure deployment options

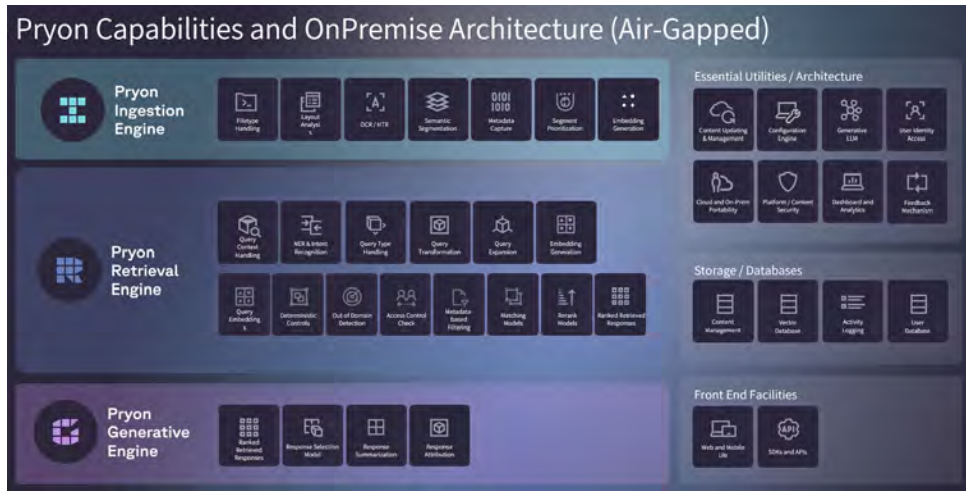


Figure: Pryon On-Premise Architecture with core capabilities, engines, utilities, storage, and databases

5. Safeguard Data Privacy with Pretrained, No-Training AI

- o *Action:* Require federal AI tools to use pretrained models that avoid customer data training, enforcing RBAC and audit logs by 2025, with privacy-by-design safeguards per the Blueprint.
- o *Rationale:* Privacy breaches undermine trust; lifecycle-secure AI supports Strategy 3, EO 14179's sovereignty focus, and Data Privacy.
- o *Impact:* Builds confidence in AI use.
- o *Proposed Policies in Action:* An SOC 2 Compliant solution, which does not use customer data to train its underlying LLM, like Pryon, mitigates risk of customer data leaks.



Figure: Pryon safeguards at Platform, Collection, and Document levels which inherit user access rights.

6. Streamline Governance with Flexible, Transparent Standards

- o *Action:* Develop a lightweight AI governance framework by 2025 that incentivizes accuracy (e.g., 95%+ retrieval rates), transparency (e.g., source attribution), and human fallback options, incorporating Algorithmic Discrimination Protections *and* Human

Alternatives.

- o *Rationale*: Overregulation hampers progress; balanced governance aligns with Strategy 6, EO 14179's deregulatory approach, and the Blueprint's equity and accountability principles.
- o *Impact*: Ensures accountability while fostering adoption.
- o *Proposed Policies in Action*: Scalable out-of-the-box COTS solutions that deliver high accuracy retrieval with source attribution, can be rapidly deployed, reducing the time to value while meeting accountability standards.



IX. Conclusion

The AI Action Plan must seize the moment to cement America's leadership by addressing inefficiencies, vulnerabilities, and regulatory missteps in today's AI landscape. By prioritizing workforce-enhancing, compute-efficient, and secure solutions the U.S. can lead responsibly, empowering its people, economy, and defenses. These recommendations offer a roadmap to a future where AI is powerful, practical, and protective of public rights, aligning with the National AI R&D Strategic Plan, EO 14179's bold vision, and the Blueprint's principles of safety, equity, privacy, transparency, and human oversight.

Pryon and Defense Industry Advisors (DIA) are uniquely positioned to drive this transformation. Pryon's RAG suite delivers secure, high-accuracy knowledge retrieval, reducing friction and aligning with *Notice and Explanation* and *Safe and Effective Systems*. DIA's expertise ensures federal readiness, supporting *Human Alternatives* in critical processes. Together, they offer a scalable, security-first approach that meets the AI Action Plan's objectives and the Blueprint's protective framework, ensuring U.S. global AI leadership.

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